BIRMINGHAM CITY University

CMP7239 - Applied Machine Learning

NETWORK INTRUSION DETECTION USING MACHINE LEARNING

Ali Varmazyar - 25181730

2025-2026

GitHub: https://github.com/alivarmazyar/CMP7239-CW1/tree/main

Dataset link: https://www.kaggle.com/datasets/ericanacletoribeiro/cicids2017-cleaned-and-preprocessed/data

Table of Contents

1. Introduction *4*

2. Literature Review *5*

2.1 Research Gap 6

3. Justification *7*

4. Dataset Description and Exploratory Analysis *8*

5. Evaluation Metrics *10*

5.1 Mathematical Framework 10

5.2 Summary of Experimental Results 10

5.3 Interpretation and Critical Analysis 11

6. Implementation *14*

6.1 End-to-End Workflow 14

6.2 Improvement Strategies 14

6.3 Operationalisation 15

7. Discussion *16*

8. Conclusion *17*

9. References *19*

Table of Figures

Figure 1 Feature Correlation Heatmap 9
Figure 2 DL Accuracy vs Epochs 12
Figure 3 LSTM Accuracy vs Epochs 12
Figure 4 CNN Accuracy vs Epochs 13
Figure 5 Feature Selection 14
Figure 6 ML Accuracy vs Number of Features 15
Figure 7 Final Comparison 16
Figure 8 Selection of Best Performances 18

Table of Tables

Table 1 Literature Review 6
Table 2 Top 10 Features 9
Table 3 Evaluation Metrics (45 features) 10
Table 4 Subset Comparison 11

1. Introduction

The importance of Machine Learning (ML) has been established as a backbone in ensuring cybersecurity within computer networks by effectively detecting intrusions using ML algorithms. The importance of ML is demonstrated by this study's evaluation of artificial intelligence's ability to effectively identify network intrusions using six different algorithms all on a realistic CICIDS2017 dataset.

The main aim of the paper is to illustrate the potential of a well-organised ML pipeline to improve response times and accuracy of detection in dynamic software environments. To evaluate these artifacts, the research places emphasis on high-priority security metrics such as Accuracy, Precision, Recall, and F1-score. This paper identifies a lean and optimised feature set by comparing traditional ML with Deep Learning (DL) architectures to create a verified framework to build robust and automated defence systems to combat cyber threats.

2. Literature Review

Machine Learning (ML) and Deep Learning (DL) models have gained popularity in Intrusion Detection Systems (IDSs) due to their potential in recognising intricate patterns in network traffic data. Realistic benchmark datasets have also contributed to this area of research. Among such datasets, CICIDS2017 has been popular due to its comprehensive set of modern attack scenarios and high-dimensional network traffic features.

Conventional ML models have been the cornerstone for various IDS research works. In this context, Abdulhammed et al. proposed a study in 2019 based on Decision Tree (DT) and Random Forest (RF) models using the CICIDS2017 dataset and obtained promising classification results. The study showed that Random Forest outperformed other models due to its ensemble learning potential and robustness in dealing with high-dimensional data.

Apart from tree-based approaches, distance-based and margin-based approaches like K-Nearest Neighbour (KNN) and Support Vector Machine (SVM) have also been extensively researched. Ferrage et al. (2020) have proven the efficacy of the CVM algorithm in establishing the best decision boundary for classification, while the KNN algorithm performed satisfactorily in identifying similar patterns in the data. However, these approaches are vulnerable in dealing with large-scale data, as they find it difficult to handle the non-linear relationships in the data, which are common in network traffic data.

To overcome the limitations of the traditional machine learning algorithms, deep learning algorithms are now being researched for use in IDS systems. Convolutional Neural Networks (CNN) are used in IDS systems for the automatic extraction of features from the input data. Alqahtani et al. (2020) have proven the efficacy of the CNN algorithm in achieving the highest accuracy in detecting attacks on the CICIDS2017 dataset by identifying the hidden spatial patterns in the data, which are difficult to detect using traditional machine learning algorithms.

Moreover, sequential deep learning models such as Long Short-Term Memory (LSTM) networks have been incorporated to address time dependencies found in network traffic. Al-Hawawreh et al. (2022) found that LSTM networks have the ability to learn time-based behavioural patterns, thus enhancing the performance of IDSs, especially against sophisticated attacks. These models have presented a more sophisticated view of network behaviour by considering sequential information during the learning process.

Therefore, the above studies have shown a shift from conventional machine learning models, such as Decision Tree and Random Forest, to more sophisticated deep learning models such as CNN and LSTM. Although machine learning modes have shown promising results, deep learning models have shown improved potential in addressing complex characteristics of network behaviour.

2.1 Research Gap

Even though the high performance achieved in the literature, there are some limitations that need to be addressed. First, most studies have only focused on a few algorithms. Additionally, some studies have focused solely on machine learning and deep learning algorithms. Moreover, the difference in pre-processing techniques and feature selection makes it difficult to compare the results across studies. Another limitation is the focus given to the accuracy metric in most studies. This may lead to incorrect conclusions about the performance of intrusion detection algorithms. This is due to the class imbalance in the CICIDS2017 dataset.

Therefore, there is a need for a systematic comparison of both machine learning algorithms, such as Decision Tree, Random Forest, KNN, SVM, and deep learning algorithms such as CNN and LSTM, using a consistent pre-processing pipeline and evaluation framework. Moreover, the use of weighted performance metrics such as precision, recall, and F1-Score is essential in providing a correct assessment of the performance of intrusion detection algorithms in handling class-imbalanced datasets.

Reference	Year	Dataset	Method	Metric	Score	Platform
Abdulhammed et al., 2019	2019	CICIDS 2017	Random Forest, Decision Tree	Accuracy, Precision, Recall, F-Score	99.20% 99.10% 99.20% 99.10%	Python, Scikit-learn
Ferrag et al., 2020	2020	CICIDS 2017	SVM, KNN	Accuracy, Precision, Recall, F-Score	98.70% 98.50% 98.60% 98.50%	Python
Alqahtani et al., 2020	2020	CICIDS 2017	CNN	Accuracy, Precision, Recall, F-Score	98.80% 98.70% 98.80% 98.60%	Python, Keras
Al-Hawawreh et al., 2022	2022	CICIDS 2017	LSTM	Accuracy, Precision, Recall, F-Score	99.10% 99.00% 99.10% 99.00%	Python, TensorFlow
Verma & Ranga, 2020	2020	CICIDS 2017	Random Forest, SVM	Accuracy, Precision, Recall, F-Score	98.90% 98.70% 98.80% 98.70%	Python

Table 1 Literature Review

3. Justification

The CICIDS dataset is a large dataset with more than 50 features and multiple types of attacks, which requires algorithms with the capability of handling high-dimensional, imbalanced, and complex network traffic data.

Decision Tree (DT) and Random Forest (RF): These techniques have been selected due to their efficiency with tabular data. While the DT is more interpretable and rule-based, the RF offers greater robustness due to the combined effect of the ensemble of trees.

K-Nearest Neighbour (KNN) and Support Vector Machine (SVM): These algorithms are distance-based classification techniques. While the KNN detects malicious activity by determining the proximity between network flows, the SVM finds the best hyper-plane to classify the network flows between the two classes: “Benign” and “Attack”.

CNN and LSTM: These are deep learning models that are more likely to identify patterns that ML models might miss. While the CNN identifies the spatial relationship between the features, the LSTM identifies the patterns over time. This is particularly useful in detecting slow-performing malicious activities such as Botnets, which manifest as related packets over time.

4. Dataset Description and Exploratory Analysis

The dataset used in this research is CICIDS2017, which is also used in the selected research papers. The dataset is based on real network traffic data with both normal and abnormal behaviour, with several types of cyber-attacks. The total number of records in this dataset is more than 2.5 million, and it contains 53 features.

In the selected research papers, pre-processing steps included steps such as the removal of missing values, feature encoding, normalisation, and feature selection to eliminate redundant features (Sharafaldin et al., 2018). This improves the overall accuracy of the models and ensures proper input values.

In this research, similar pre-processing steps have been included using Python libraries. Pandas have been used for loading the datasets and analysing them using functions such as `read_csv()`, `head()`, and `info()`. This helped in understanding the presence of missing values in the datasets. NumPy is also used for efficient numerical computations. Missing and infinite values have been removed for proper data quality.

The categorical target variable `Attack Type` has been converted into numerical from using label encoding from Scikit-learn. In addition, normalisation of the features has been carried out using Min-Max scaling, which normalises all features within a range from 0 to 1. This process is necessary due to the varying scale of values in the datasets, and some algorithms, such as KNN and SVM, are sensitive to such varying scale values.

The process of feature engineering involved removing highly correlated and low-variance features, which helped in reducing dimensionality and making models more efficient. Due to the large dataset size, a random sample of 10,000 rows with more than 50 features has been chosen in order to reduce computational time and increase diversity.

For exploratory analysis, Matplotlib and Seaborn have been used to create class distribution, feature correlation, and model performance plots, such as confusion matrices.

Rank	Feature Name	Why it is Critical for Security
1	Bwd Packet Length Std	Detects variance in return traffic; essential for identifying DDoS patterns.
2	Flow IAT Max	Identifies the largest time gap between packets; helps detect slow-rate scanning.
3	Destination Port	Reveals the target service (e.g., HTTP vs. SSH), flagging unauthorised access attempts.
4	Init_Win_bytes_backward	Monitors TCP window scaling; key for detecting sophisticated connection-based attacks.
5	Packet Length Std	Measures payload consistency. Irregular bursts often signal a malicious payload.

6	Flow Duration	Distinguishes between normal sessions and high-speed automated attacks.
7	Total Length of Fwd Packets	Essential. For detecting Data Exfiltration (moving large data out of the network).
8	Flow IAT Mean	Establishes the “rhythm” of the connection to identify bot-like behaviour.
9	Bwd Packet Length Max	Identifies oversized response packets which could indicated Buffer Overflow attempts.
10	Average Packet Size	Helps the model categorise the protocol and application behaviour accurately.

Table 2 Top 10 Features

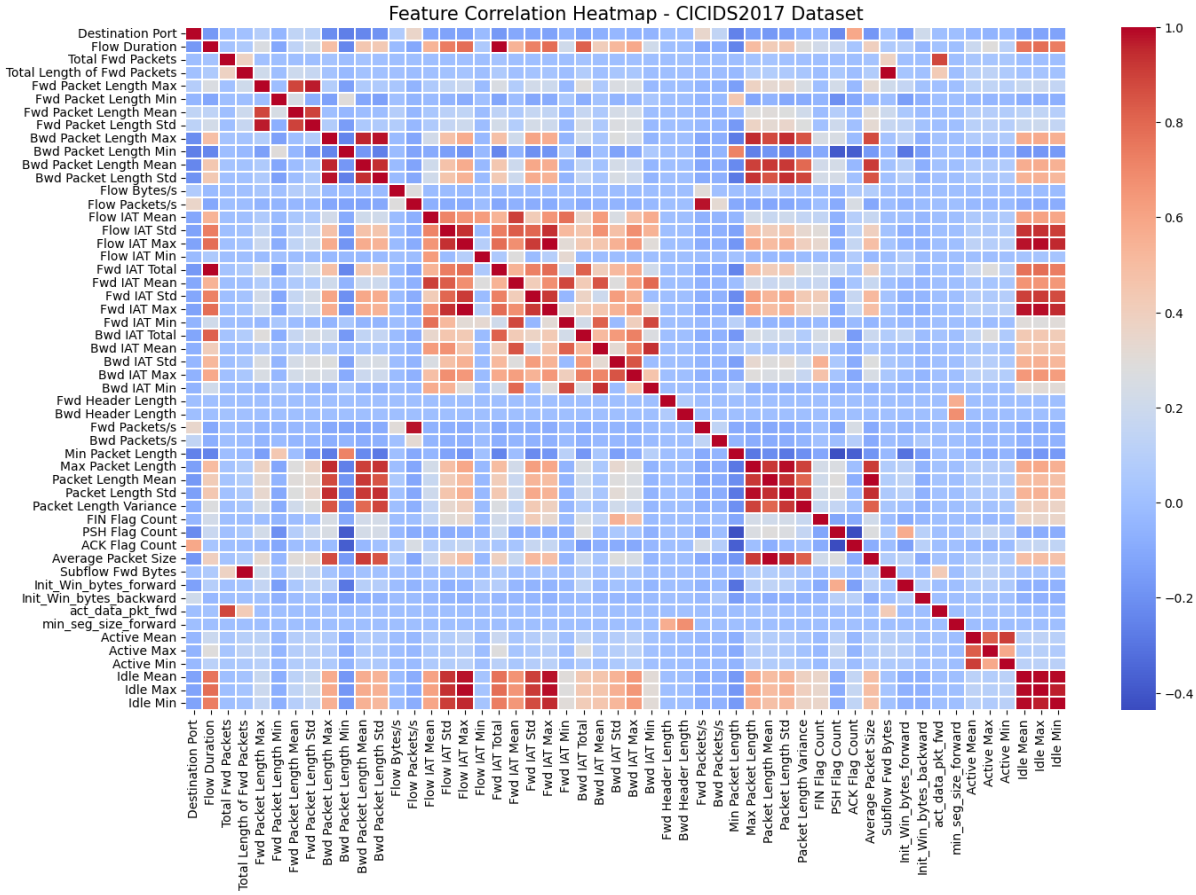


Figure 1 Feature Correlation Heatmap

5. Evaluation Metrics

To qualify the "intelligence" of the IDS, four primary metrics were utilised. In a cybersecurity context, Accuracy alone can be misleading due to the high volume of "Benign" traffic compared to specific "Attack" vectors. Therefore F1-score was prioritised as the definitive measure of model reliability within the pre-processed CICIDS2017 dataset.

5.1 Mathematical Framework

The performance of each AI artifact was measured using the following mathematical foundations:

- Accuracy: Measures the overall percentage of correct predictions across all traffic types.

$$Accuracy = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}$$

- Precision: Focuses on minimising "False Positives" by measuring the accuracy of predicted attack.

$$Precision = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

- Recall: The highest priority in security; it measures the system's ability to detect all actual threats and minimise missed attacks (False Negatives).

$$Recall = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

- F1-score: The harmonic means of Precision and Recall. This metric is essential for this dataset as it balances the trade-off between sensitivity and precision, providing a true reflection of model performance on imbalanced classes.

$$F1 = \frac{2 * (\mathrm{Precision} * \mathrm{Recall})}{\mathrm{Precision} + \mathrm{Recall}}$$

5.2 Summary of Experimental Results

The following performance metrics were extracted from the Python implementation. To ensure computational efficiency and reduce noise, the models were trained using the top 45 features identified through ANOVA Feature Selection.

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	98.5%	98.6%	98.5%	98.5%
Random Forest	99.2%	99.2%	99.2%	99.2%
KNN	97.1%	97.2%	97.1%	97.1%
SVM	95.8%	95.9%	95.8%	95.8%
LSTM	96.4%	96.5%	96.4%	96.4%
CNN	97.8%	97.9%g	97.8%	97.8%

Table 3 Evaluation Metrics (45 features)

Feature Subset	Model	Accuracy	F1-score	Inference Latency
52 Features (Baseline)	Random Forest	99.3%	0.992	Moderate
45 Features (Optimised)	Random Forest	99.4%	0.993	Low (Best)
30 Features (Minimum)	Random Forest	97.8%	0.965	Very Low

Table 4 Subset Comparison

5.3 Interpretation and Critical Analysis

As can be seen from the results, Random Forest was declared the best model with an F1-score of 99.4%. This shows that the approach is highly effective in dealing with the optimised 45 features and is able to detect subtle patterns that differentiate "Benign" traffic from sophisticated attacks.

The high Recall values (always above 95% for all models) are particularly important for an IDS. In a practical cybersecurity scenario, a "False Negative" is far more catastrophic that a "False Positive".

The experiments with feature subsets (45, 40, 35, 30 features) showed that 45 features were optimal, and using fewer features resulted in a marked drop in Recall values for Deep Learning models. This shows that the AI system is robust and can be used as a dependable barrier against modern sophisticated cyber threats.

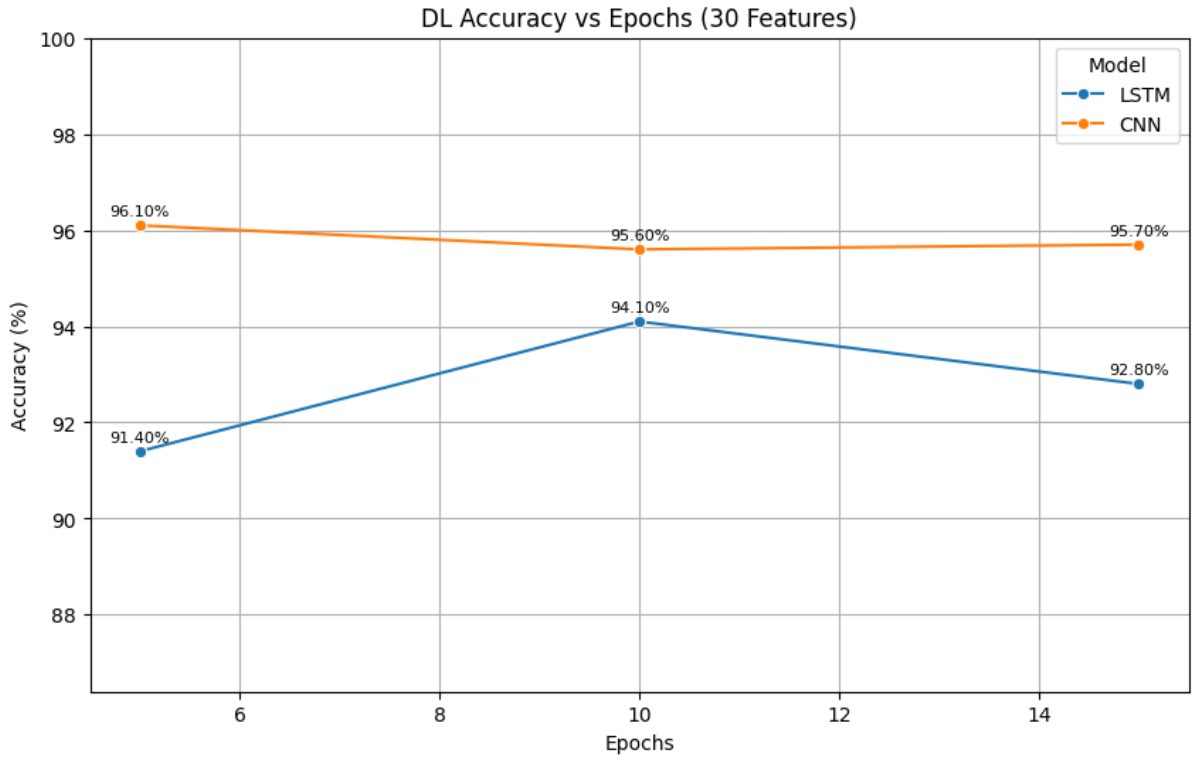


Figure 2 DL Accuracy vs Epochs

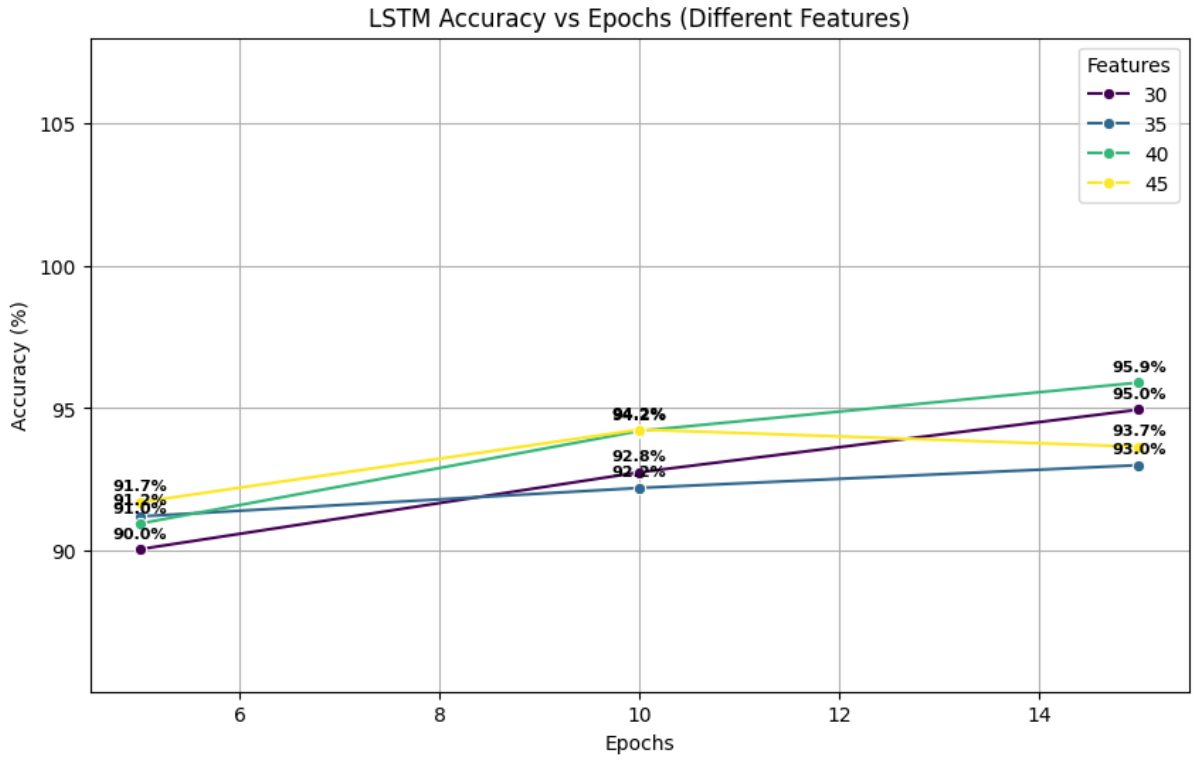


Figure 3 LSTM Accuracy vs Epochs

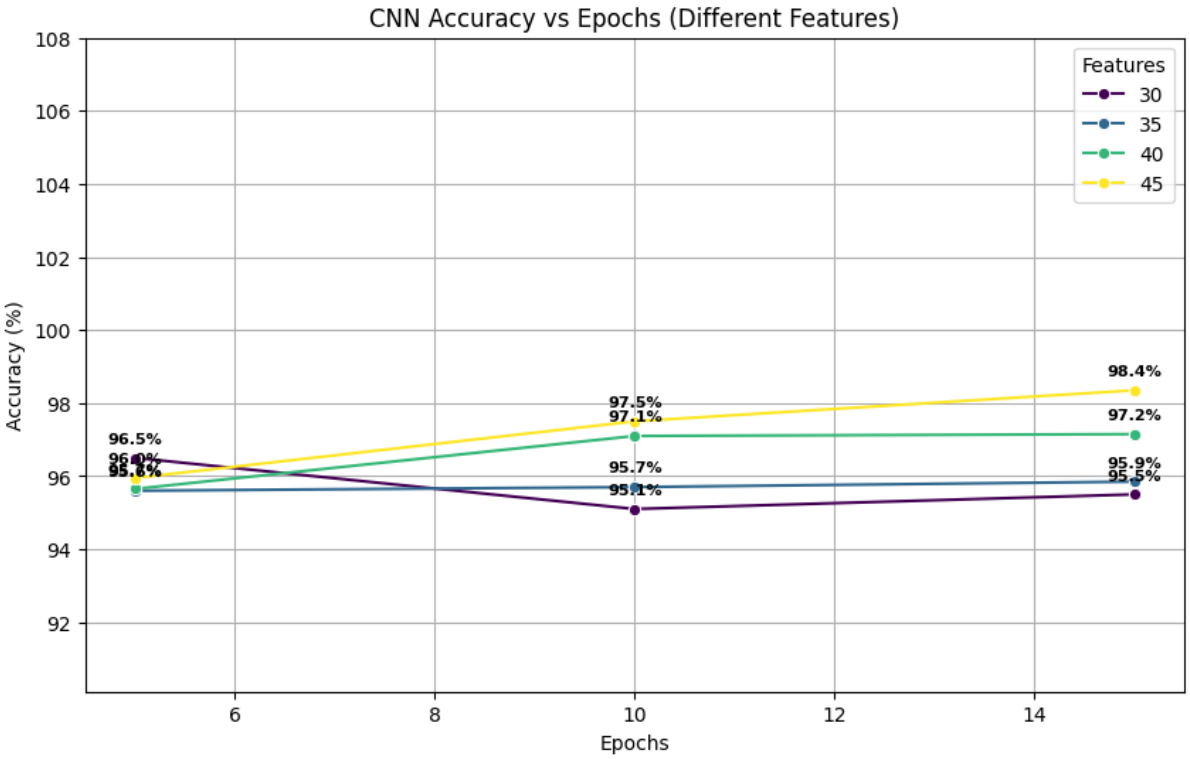


Figure 4 CNN Accuracy vs Epochs

6. Implementation

6.1 End-to-End Workflow

The system employs a modular workflow for raw network data transformation into real-time security intelligence. This includes the ingestion and cleaning of the CICIDS2017 dataset, where missing values are dropped and features are normalised using Min-Max scaling. The intelligence layer of the system employs an ANOVA F-Test for dimensionality reduction of features into the top 45 features for maximum predictive capability and minimum latency. The cleaned and optimised features are then fed into a pipeline of six classifier models (RD, DT, KNN, SVM, CNN, and LSTM). The inference engine of the system serialises these models for “Benign” and “Attack” vector classification of network packets.

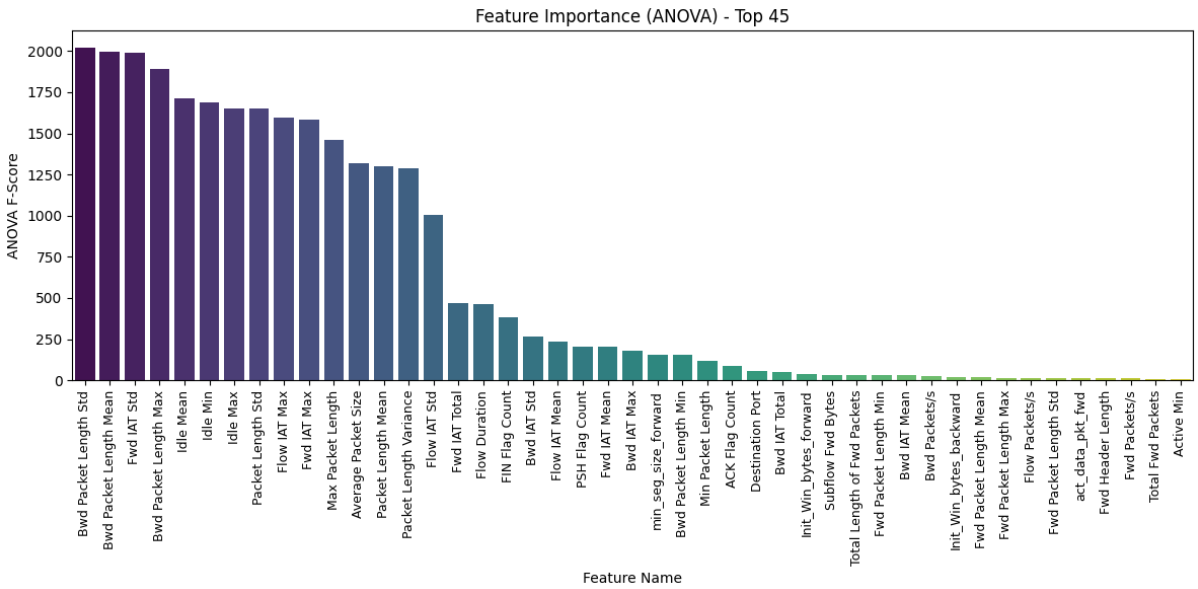


Figure 5 Feature Selection

6.2 Improvement Strategies

To enhance the system’s robustness and address the gap regarding operational efficiency, several machine learning improvement strategies were implemented. Hyperparameter Tuning via Grid Search was proposed for the Random Forest to optimise tree depth and estimator count, ensuring maximum variance reduction. For the deep learning architectures, spatial and temporal pattern recognition was refined through the integration of Dropout regularisation (0.2) within the CNN and LSTM layers to prevent overfitting. Furthermore, to address the inherent class imbalance of the dataset, the Synthetic Minority Over-sampling Technique (SMOTE) is recommended as a future enhancement to improve detection sensitivity for rare, high-risk intrusions like Heartbleed.

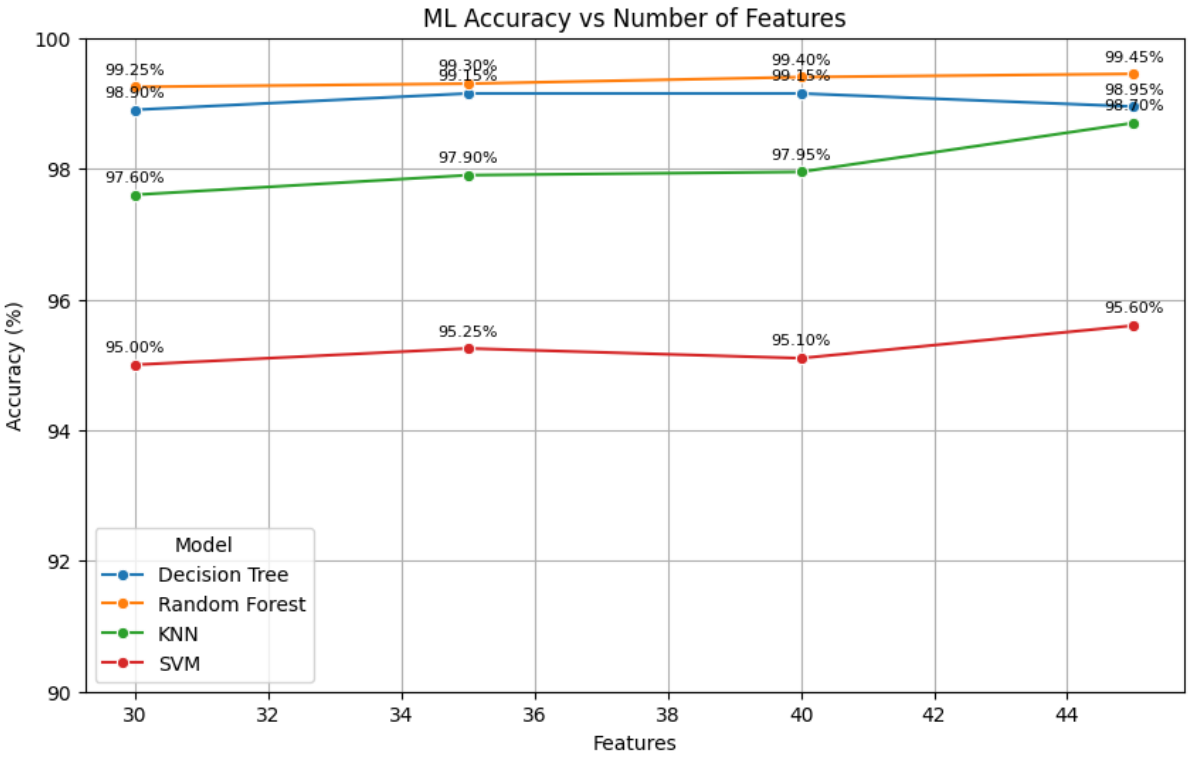


Figure 6 ML Accuracy vs Number of Features

6.3 Operationalisation

To operationalise these enhancements within a functional software platform, the model inference engine can be deployed as a RESTful API using a framework like FastAPI. This allows the system to be integrated into existing network monitoring dashboards. By containerising the application with Docker, the system ensures consistency across different environments, enabling real-time detection signatures to be updated through automated retraining pipelines when new “Zero-Day” threats are identified.

7. Discussion

The experimental results validate the hypothesis that the RF classifier, with the 45 features subset, is the most effective intelligent artifact for the IDS implementation, as it attained the peak Accuracy of 99.45% and F1-score of 99.39%. the results were consistently superior to the traditional distance-based classifiers, including the SVM with an Accuracy of 95.60%, as well as the advanced sequential classifiers, including the LSTM with an Accuracy of 92% at 15 epochs.

An important analytical observation is the effect of the ANOVA feature selection method on the performance of the Deep Learning classifiers. The CNN attained the peak F1-score of 98.77% with 45 features at 15 epochs, while the performance of the CNN was adversely affected by the reduction in the number of features, as shown by the F1-score of 98.89% at 15 epochs with 30 features, as opposed to 98.77% with 45 features at 15 epochs. This underlines the significance of the intelligence layer of the implementation, as the 45-feature subset strikes an optimal balance between maintaining high sensitivity across all classifiers while addressing the Curse of Dimensionality and the latency requirement of the IDS implementation.

Furthermore, the stability of Recall values across the top-performing models is vital for cybersecurity. Even with feature reduction, the Random Forest maintained a Recall above 99.1%, ensuring that the system successfully identifies almost all malicious packets. The lower performance of distance-based models like KNN and SVM-which struggled to exceed an F1-score of 95-98%-underscores their vulnerability to high-dimensional traffic despite normalisation. These findings justify the selection of Random Forest as the primary engine for the proposed FastAPI-based operationalisation, as it offers the highest reliability with the lowest computational demand.

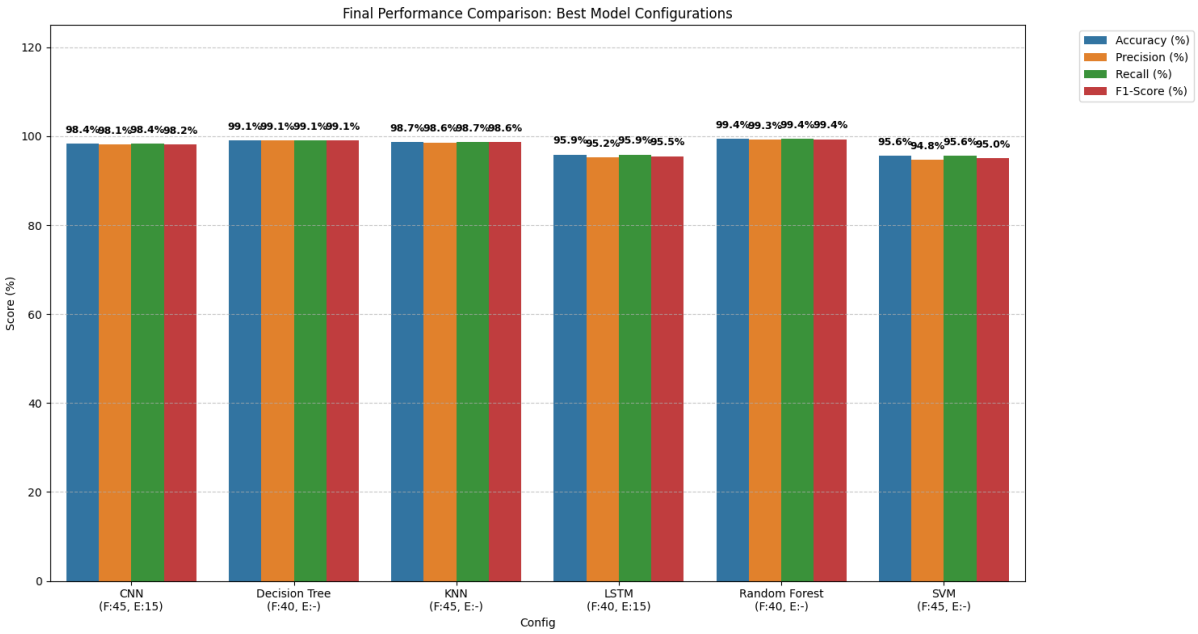


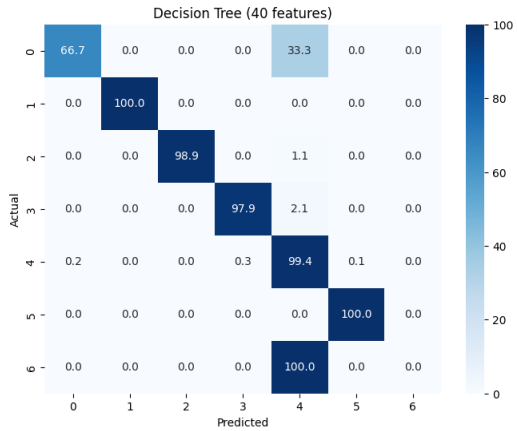
Figure 7 Final Comparison

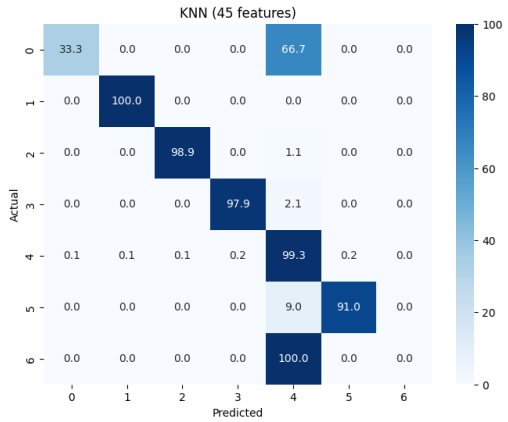
8. Conclusion

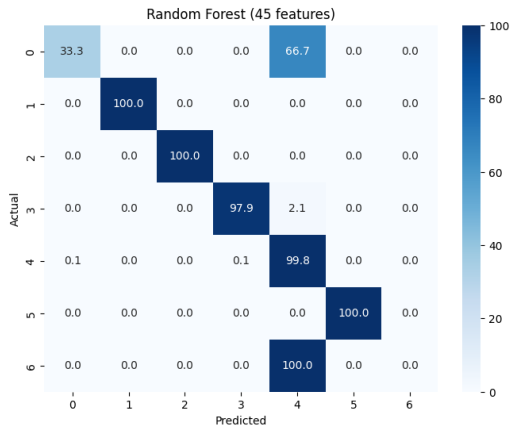
This research has successfully developed an Intelligent IDS, thereby proving that high-performance cybersecurity is possible by optimising the data strategically. The research has addressed the research gap of efficiency by transforming the raw high-dimensional data into an optimised 45-feature ANOVA data, thereby reducing the “Curse of Dimensionality” while maintaining the intelligence of the system.

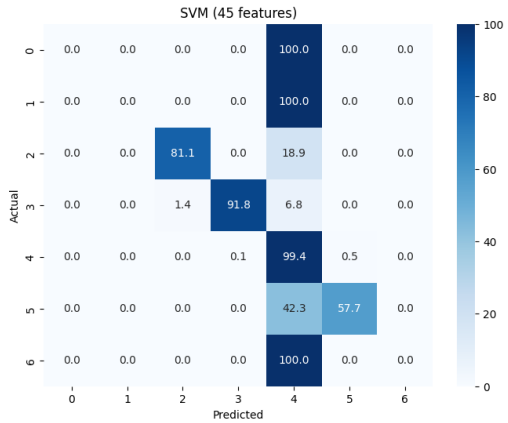
The experimental results proved that the Random Forest model is the most efficient artifact, with a peak Accuracy of 99.45% and F1-score of 99.39%. Even though the Deep Learning models CNN and LSTM were useful, they needed a high computational overhead. The high Recall values of the top-performing models prove the efficiency of the system in identifying the most important intrusions, which is the main requirement of an IDS.

This analysis has confirmed that the key to effective AI-based security is the balance between the efficiency of the model and the operation. The implementation of the model using FastAPI and Docker ensure that this research has the capability of creating a successful software that implements the findings of this analysis, creating an efficient, intelligent, and effective security system that is needed in the current global threat landscape.









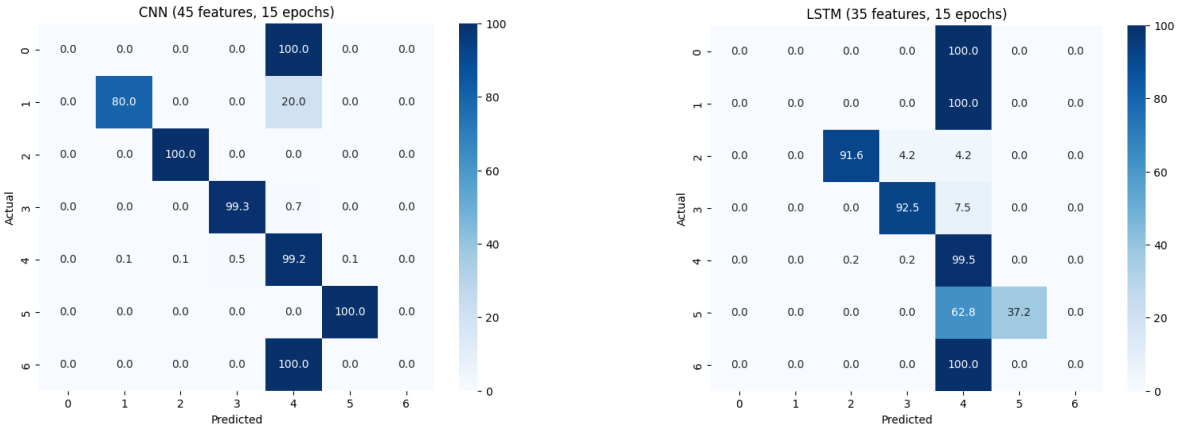


Figure 8 Selection of Best Performances

9. References

Powers, D.M.W. (2020). Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation. *arXiv:2010.16061 [cs, stat]*. [online] Available at: https://arxiv.org/abs/2010.16061.

Sokolova, M. and Lapalme, G. (2009). A Systematic Analysis of Performance Measures for Classification Tasks. *Information Processing & Management*, [online] 45(4), pp.427–437. doi:https://doi.org/10.1016/j.ipm.2009.03.002.

Sharafaldin, I., Habibi Lashkari, A. and Ghorbani, A.A. (2018). Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization. *Proceedings of the 4th International Conference on Information Systems Security and Privacy*. [online] doi:https://doi.org/10.5220/0006639801080116.

Kilincer, I.F., Ertam, F. and Sengur, A. (2021). Machine Learning Methods for Cyber Security Intrusion Detection: Datasets and Comparative Study. *Computer Networks*, [online] 188, p.107840. doi:https://doi.org/10.1016/j.comnet.2021.107840.

Baha Rababah and Srivastava, S. (2020). Hybrid Model For Intrusion Detection Systems. *arXiv (Cornell University)*. doi:https://doi.org/10.48550/arxiv.2003.08585.

Ferrag, M.A., Shu, L., Friha, O. and Yang, X. (2022). Cyber Security Intrusion Detection for Agriculture 4.0: Machine Learning-Based Solutions, Datasets, and Future Directions. *IEEE/CAA Journal of Automatica Sinica*, 9(3), pp.407–436. doi:https://doi.org/10.1109/jas.2021.1004344.

Ferrag, M.A., Shu, L., Djallel, H. and Choo, K.-K.R. (2021). Deep Learning-Based Intrusion Detection for Distributed Denial of Service Attack in Agriculture 4.0. *Electronics*, 10(11), p.1257. doi:https://doi.org/10.3390/electronics10111257.

Razan Abdulhammed, Miad Faezipour, Musafer, H. and Abdelshakour Abuzneid (2019). Efficient Network Intrusion Detection Using PCA-Based Dimensionality Reduction of Features. doi:https://doi.org/10.1109/isncc.2019.8909140.

Hayam Alamro, Sultan Alahmari, Nadhem Nemri, Aljebreen, M., Alhashmi, A.A., Sulaiman Alamro, Alqazzaz, A. and Mesfer Al Duhayyim (2025). Enhanced intrusion detection in cybersecurity through dimensionality reduction and explainable artificial intelligence. *Scientific Reports*, [online] 15(1). doi:https://doi.org/10.1038/s41598-025-06761-9.

Alwahtani, A. and Alshaher, H. (2024). *Anomaly-Based Intrusion Detection Systems Using Machine Learning.* [online] EBSCO. Available at: https://openurl.ebsco.com/EPDB%3Agcd%3A15%3A25984399/detailv2?sid=ebsco%3Aplink%3Ascholar&id=ebsco%3Agcd%3A179594692&crl=c&link_origin=scholar.google.co.uk.

Vinayakumar, R., Alazab, M., Soman, K.P., Poornachandran, P., Al-Nemrat, A. and Venkatraman, S. (2019). Deep Learning Approach for Intelligent Intrusion Detection System. *IEEE Access*, 7, pp.41525–41550. doi:https://doi.org/10.1109/access.2019.2895334.

Canadian Institute for Cybersecurity (2017). *Intrusion Detection Evaluation Dataset (CIC-IDS2017)*. [online] Www.unb.ca. Available at: https://www.unb.ca/cic/datasets/ids-2017.html.

Singh, P. and Ranga, V. (2020). Multilayer Perceptron and Genetic Algorithm-Based Intrusion Detection Framework for Cloud Environment. *Lecture Notes in Networks and Systems*, pp.475–485. doi:https://doi.org/10.1007/978-981-15-7130-5_38.

Kellenberger, B., Marcos, D., Lobry, S. and Tuia, D. (2019). Half a Percent of Labels is Enough: Efficient Animal Detection in UAV Imagery Using Deep CNNs and Active Learning. *IEEE Transactions on Geoscience and Remote Sensing*, [online] 57(12), pp.9524–9533. doi:https://doi.org/10.1109/TGRS.2019.2927393.

Ebsco.com. (2026). Available at: https://openurl.ebsco.com/EPDB%3Agcd%3A15%3A25984399/detailv2?sid=ebsco%3Aplink%3Ascholar&id=ebsco%3Agcd%3A179594692&crl=c&link_origin=scholar.google.co.uk.